

The Revit Plugin

Documentation

<i>Version</i>	<i>Date</i>	<i>Author</i>	<i>Changes/Notes</i>
1.0	15.07.2025	Yehia Abdelazziz	The original version contained the two plugin tools(Hide Elements for isolating specific elements and Automatic parameter insertion tool). This version added the Check For Updates tool with explanation and included the Automatic Installation section for flawless and quick Revit.Addin.2024 tool installation

The Revit Plugin Documentation

Warning

Please **DO NOT** attempt to modify any data in the Excel sheet outside a Revit session to avoid faulty behavior from the update functionality of the plugin.

Plugin Installation

Automatic Installation

In the compressed .zip file (Revit.Addin.2024.zip) one can find the following files: a folder (Revit.Adding.2024), Installer.bat, ParameterData.xlsx (Our main data-source) and this document ReadMeFirst.pdf. The following [Figure](#) shows the mentioned content above.

 Revit.Addin.2024	15.07.2025 10:09	Dateiordner	
 Installer.bat	14.07.2025 14:48	Windows-Batchda...	2 KB
 ParameterData.xlsx	07.07.2025 15:34	Microsoft Excel-Ar...	21 KB
 ReadMeFirst.pdf	10.07.2025 12:59	Adobe Acrobat-D...	4.525 KB

Figure 1 .zip contents

Extract the .zip file anywhere and save the ParameterData.xlsx somewhere known. Proceed to double-click on the Installer.bat which initiates the installation process. A PowerShell window will pop up prompting the user for a (J/N) to unblock the blocked .dll files, simply press J (Ja). The following [Figure](#) depicts this.

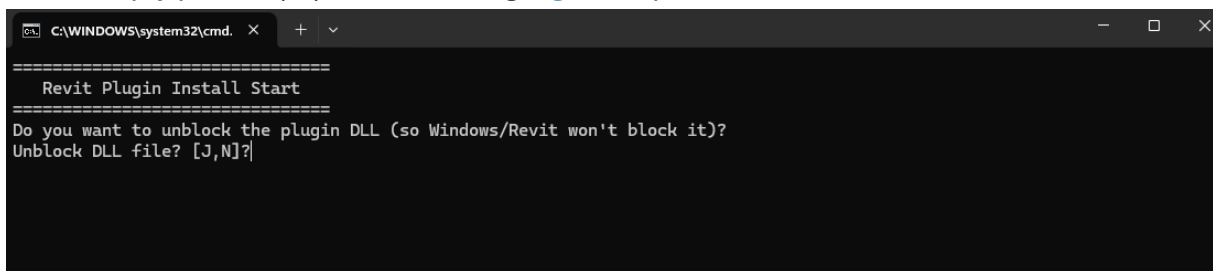


Figure 2 Installation Window

After pressing J, the installation begins by adding the necessary files into the Revit addin directory. The messages displayed when the installation is finished should be similar to the ones in the follow [Figure](#).

```
C:\WINDOWS\system32\cmd. X + v

=====
Revit Plugin Install Start
=====
Do you want to unblock the plugin DLL (so Windows/Revit won't block it)?
Unblock DLL file? [J,N]?J
Unblocking...
Copying DLLs...
79 Datei(en) kopiert
Copying .addin file...
1 Datei(en) kopiert

Plugin installation was successful.
Drücken Sie eine beliebige Taste . . . |
```

Figure 3 Successful Installation

You can then close the window and do a quick manual check by going to this URI `C:\ProgramData\Autodesk\Revit\Addins\2024`. Simply insert this into any folder explorer and look for the following file: `Revit.Addin.2024.addin` and the following folder: `PluginTrail`. If you have successfully located the folder and the file just as shown in the [Figure](#) below then proceed to this section [Post-Installation](#).

Name	Änderungsdatum	Typ	Größe
ExportViewSelectorAddin	16.04.2025 14:47	Dateiordner	
Autodesk.BatchPrint.addin	16.04.2025 14:50	ADDIN-Datei	1 KB
Autodesk.Collaborate.addin	16.04.2025 14:47	ADDIN-Datei	2 KB
Autodesk.eTransmitApplication.addin	16.04.2025 14:49	ADDIN-Datei	1 KB
Autodesk.WorksharingMonitor.Applicati...	16.04.2025 14:50	ADDIN-Datei	2 KB
ExportViewSelectorApp.addin	11.09.2023 12:41	ADDIN-Datei	1 KB
FormItConverter.addin	16.04.2025 14:49	ADDIN-Datei	2 KB
PluginTrail	15.07.2025 15:45	Dateiordner	
Revit.Addin.2024.addin	10.07.2025 17:01	ADDIN-Datei	1 KB

Figure 4 Files Installed

In case something went wrong, go to the [Manual Installation](#) section and follow the instructions. If the problem persists, contact the developer.

Manual Installation

In the compressed .zip files (`Revit.Addin.2024.zip`) an excel sheet (`ParameterData.xlsx`) and a folder (`Revit.Addin.2024`) should exist as in [Figure 1](#). The excel file is what we will call the data source, which holds the information about the parameters and their corresponding groups along with the values. Screenshots will be provided for better illustration.

Copy the excel file to a known path and proceed to copy the contents inside the Revit.Addin.2024 folder in this path on your computer (the .addin file might not have an icon, which is fine). The steps are illustrated in the following figures.

Name	Typ
PluginTrail	Dateiordner
Revit.Addin.2024.addin	ADDIN-Datei

Figure 5 Contents of Folder Revit.Addin.2024

The contents in [Figure 2](#) should be copied in this path first `C:\ProgramData\Autodesk\Revit\Addins\2024`

Note: ProgramData is sometimes hidden. To unhide the folder, go to C then in the task bar click on **Anzeigen > Einblenden > Ausgeblendete Elemente**.

Once you have located this folder, you should be inside the 2024 folder. Now paste the PluginTrail folder and the Revit.Addin.2024.addin file in the 2024 folder as shown below in [Figure 3](#).

ProgramData > Autodesk > Revit > Addins > 2024 >				
2024 durchsuchen				
Sortieren Anzeigen				
Name	Änderungsdatum	Typ	Größe	
ExportViewSelectorAddin	16.04.2025 14:47	Dateiordner		
PluginTrail	03.07.2025 11:49	Dateiordner		
Autodesk.BatchPrint.addin	16.04.2025 14:50	ADDIN-Datei	1 KB	
Autodesk.Collaborate.addin	16.04.2025 14:47	ADDIN-Datei	2 KB	
Autodesk.eTransmitApplication.addin	16.04.2025 14:49	ADDIN-Datei	1 KB	
Autodesk.WorksharingMonitor.Applicati...	16.04.2025 14:50	ADDIN-Datei	2 KB	
ExportViewSelectorApp.addin	11.09.2023 12:41	ADDIN-Datei	1 KB	
FormItConverter.addin	16.04.2025 14:49	ADDIN-Datei	2 KB	
Revit.Addin.2024.addin	07.07.2025 14:47	ADDIN-Datei	1 KB	

Figure 6 Plugin Path contents

Afterwards, navigate inside the PluginTrail folder and locate this .dll file, Revit.Addin.2024.dll. Shown in [Figure 4](#) below.

RBush.dll	18.11.2024 16:48
RebarDBAPI.dll	13.09.2023 22:58
Revit.Addin.2024.dll	09.07.2025 10:36
RevitAddInUtility.dll	13.09.2023 22:58

Figure 7 The plugin .dll file

Microsoft ensures the safety of your computer by blocking .dll files because they can intervene with your computer if they're created by a harmful person. In our case we only

created the .dll for Revit plugin duties so we need to unblock the file. Right click on the Revit.Addin.2024.dll file press on **Weitere Optionen anzeigen > Eigenschaften** then a window will appear, shown in [Figure 5](#).

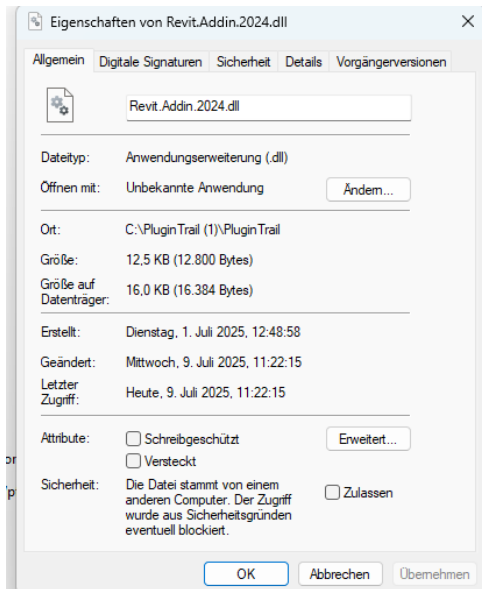


Figure 8 Setting Window

At bottom of the window, you will see a note beside **Sicherheit**. You need to check the **Zulassen** box beside the note message as shown below in [Figure 6](#).

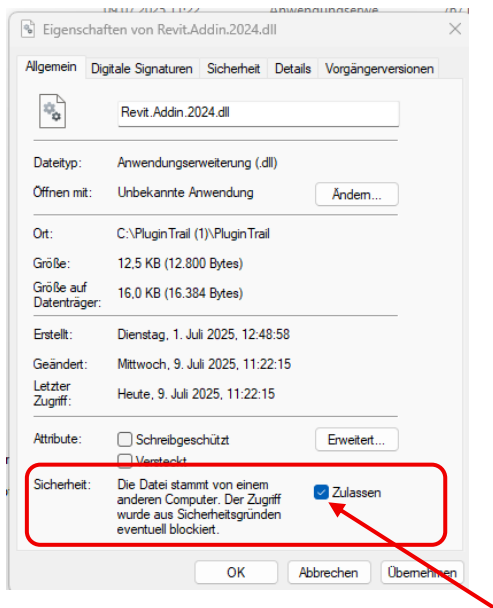


Figure 9 ZuLassen Checked

Now that we have finished this part, we will proceed to copy the files to the second directory. The same files are to be copied PluginTrail folder and the Revit.Addin.2024.addin file as shown below in [Figure 7](#).

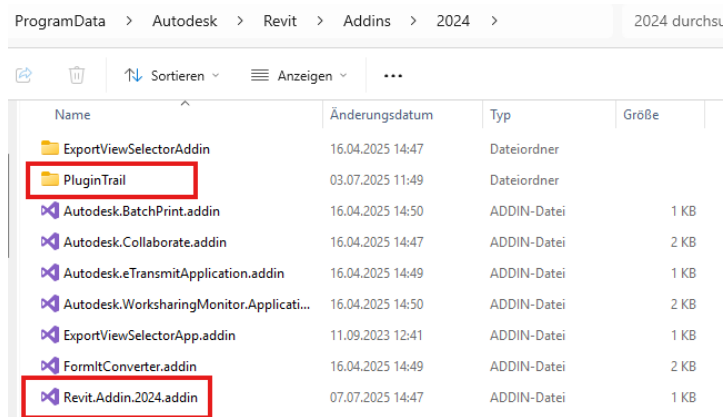


Figure 10 Files to be copied

Please note that the following step is only necessary if the plugin did not load successfully or resulted in a .dll error upon usage. Navigate to the URI of your explorer as shown in [Figure 8](#) and proceed to write this %appdata% and press enter.

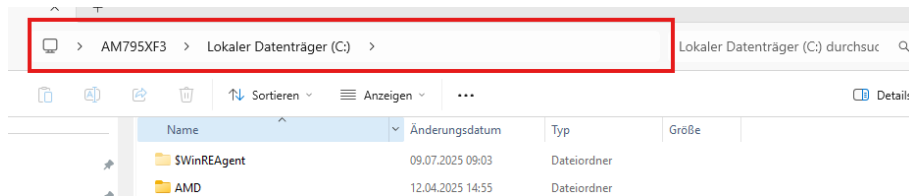


Figure 11 Explorer URI

Navigate from here to **Autodesk > Revit > Addin > 2024** then paste the copied content into this folder. The path should be similar to this, with a slight difference to the username of your computer.

`C:\Users\YourUserName\AppData\Roaming\Autodesk\Revit\Addins\2024`

YourUserName will be replaced with the user name that you are logged in with. If all done correctly, you should have a folder structure like the one shown in [Figure 9](#) below.

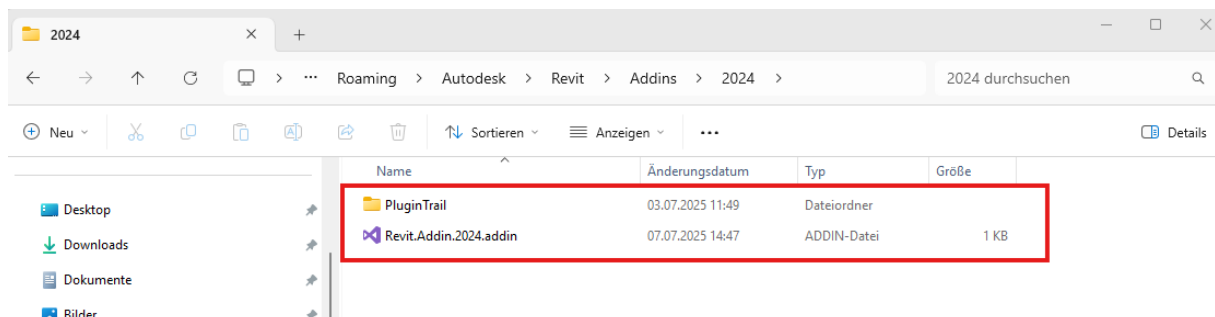


Figure 12 Contents in the second directory

We will make a small modification to the Revit.Addin.2024.addin file to include the path of this directory, we haven't done this in the previous one because the path is always similar from computer to another, but this directory is different.

Open the .addin with a notepad you should see something like this in [Figure 10](#).

```
<?xml version="1.0" encoding="utf-8" standalone="no"?>
<RevitAddIns>
  <AddIn Type="Application">
    <Name>Plugin Tools</Name>
    <Assembly>"C:\ProgramData\Autodesk\Revit\Addins\2024\PluginTrail
\Revit.Addin.2024.dll"</Assembly>
    <AddInId>1e1d58d1-c8a1-4f1d-8c36-bef98a3773dc</AddInId>
    <FullClassName>Revit.Addin._2024.App</FullClassName>
    <VendorId>YH07</VendorId>
    <VendorDescription>Müller+Hereth</VendorDescription>
  </AddIn>
</RevitAddIns>
```

Figure 13 The Revit.Addin.2024.addin File

Then proceed to navigate inside the PluginTrail folder and locate the Revit.Addin.2024.dll file as shown below in [Figure 11](#).

	RBush.dll	18.11.2024 16:48
	RebarDBAPI.dll	13.09.2023 22:58
	Revit.Addin.2024.dll	09.07.2025 10:36
	RevitAddInUtility.dll	13.09.2023 22:58

Figure 14 Contents inside PluginTrail Folder

Right click the Revit.Addin.2024.dll file and choose **Als Pfad kopieren** shown in [Figure 12](#).

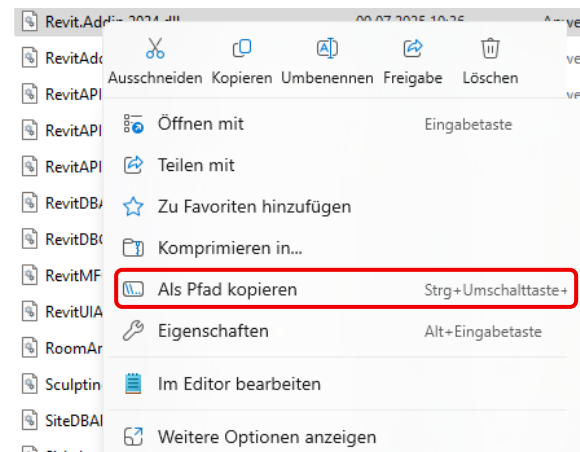


Figure 15 Als Pfad kopieren

Afterwards, we'll paste the path into the Revit.Addin.2024.addin file that we opened with a notepad. [Figure 13](#) shows where you paste the path.

```
<?xml version="1.0" encoding="utf-8" standalone="no"?>
<RevitAddIns>
  <AddIn Type="Application">
    <Name>Plugin Tools</Name>
    <Assembly>"C:\Users\Praktikant.Frlg\AppData\Roaming\Autodesk\Revit\Addins\2024
    \PluginTrail\Revit.Addin.2024.dll"</Assembly>
    <AddInId>1e1d58d1-c8a1-4f1d-8c36-bef98a3773dc</AddInId>
    <FullClassName>Revit.Addin._2024.App</FullClassName>
    <VendorId>YH07</VendorId>
    <VendorDescription>Müller+Hereth</VendorDescription>
  </AddIn>
</RevitAddIns>
```

Figure 16 the .addin file after pasting the path

The path that you have copied should be pasted and replaces the highlighted one as shown in this section between the **<Assembly>** "C:\Users\Praktikant.Frlg\AppData\Roaming\Autodesk\Revit\Addins\2024\PluginTrail\Revit.Addin.2024.dll" **</Assembly>** tags.

Save and close the file.

Post-installation

Open Revit 2024, and if everything has been done correctly you should receive this message in [Figure 14](#).

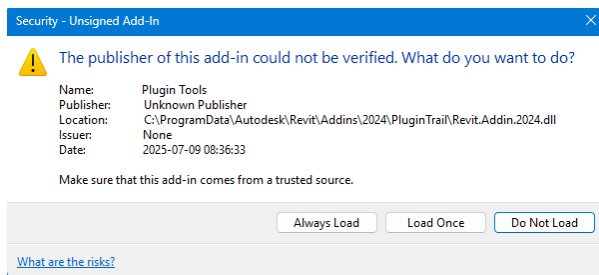


Figure 17 Warning message of Addins

The warning message appears because our Addin is not certified by Microsoft, which is a program that Autodesk and Microsoft initiated for their softwares, we will choose Always load (Immer Laden). Continue checking; by navigating to Ribbon above and searching for a tab named “My Tools” click on and you should see the three buttons for the plugins we just installed shown in [Figure 15](#). This means that we installed the plugins correctly and Revit sees and identifies them.

In the next section, we will explain how to use the plugins and at the same time check if the behavior is correct.

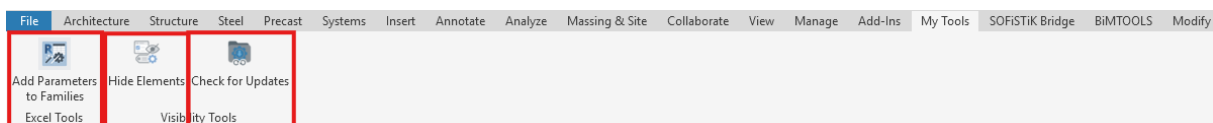


Figure 18 The three plugins

How To Use The Plugins

Hide Elements Plugin

The plugins logic depends on the parameter **Bauteilnummer** to uniquely identify the **Bauteil** in the project. For instance, the bodenplatte, Wand, Bohrpfaahl etc..

Press on the Hide Elements tool a window will pop up that has a list of checkboxes each box corresponds to all the elements that have this Bauteilnummer beside the box as shown in [Figure 16](#). The window also includes two buttons, **Isolate** and **UnHide All**.

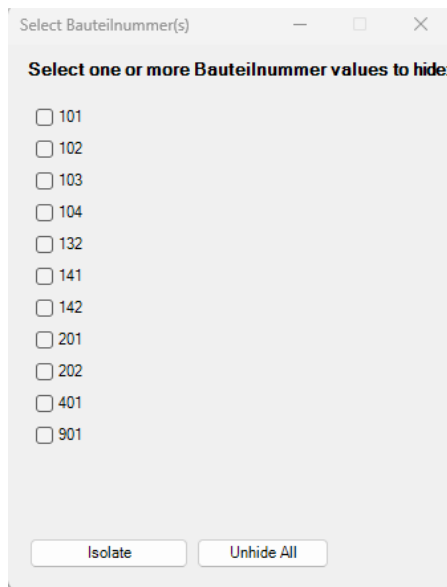


Figure 19 Hide Elements Window

Choose for example the numbers 101 and 142 to demonstrate the functionality and press Isolate. This will create a temporary view that hides everything except elements that have Bauteilnummer 101 and 142.

- If you wish to revert this, you can open the tool again and press the Unhide button.
- If you wish to add more elements to the temporary view, you need to Unhide everything first and then choose the desired elements to isolate in a temporary view

[Figure 17](#) shows an old project to demonstrate the functionality and [Figure 18](#) depicts the choosing of Bauteilnummers 101 and 142 while [Figure 19](#) shows the result inside the Revit project.

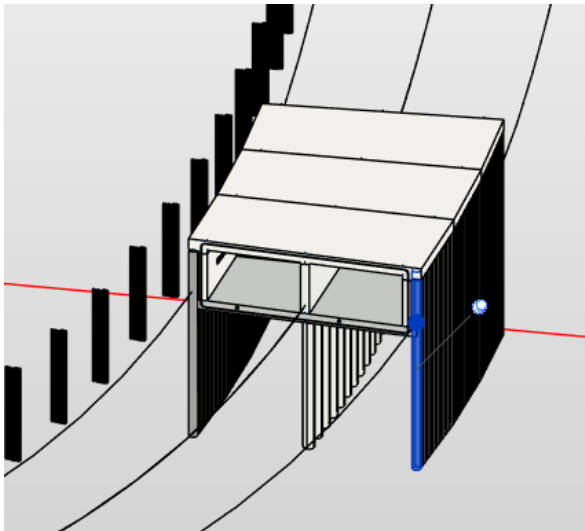


Figure 20 Old Project Model

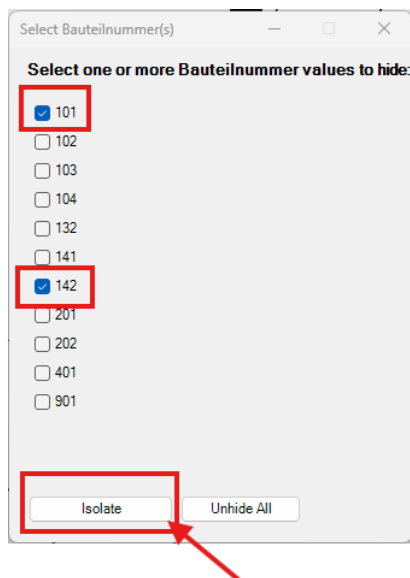


Figure 21 Choosing Bauteilnummers and pressing Isolate

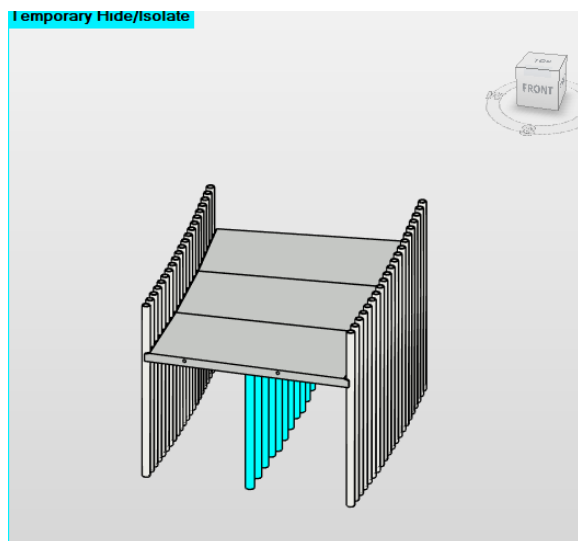


Figure 22 The Result of Choosing Bauteilnummers 101 and 142

This will pop up a window as shown in [Figure 22](#). The **Create** button that we will press on *(Please make sure that you remember the path to your original shared parameters file .txt so that you can choose it after this demonstration is finished)* and save into a directory or path known to us to be able to locate later. When doing so, Revit will understand that we want to use this list as our default for shared parameters.

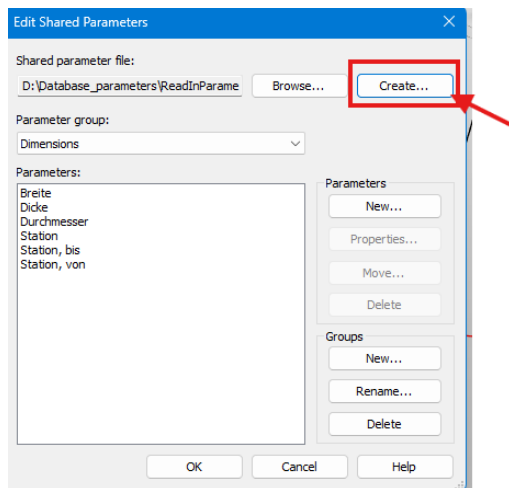


Figure 25 Edit Shared Parameters Window

After creating a new list, the window should look something like this in [Figure 23](#) completely empty list.

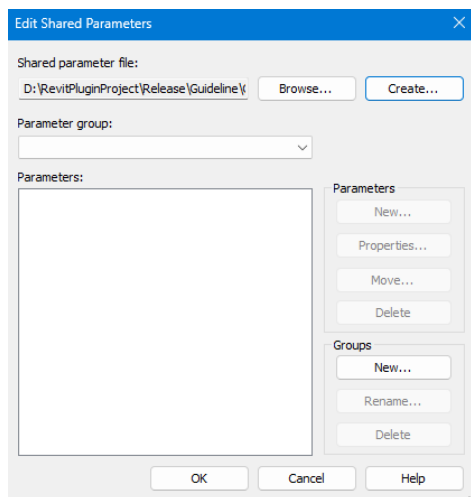


Figure 26 Empty list

Step 2

The previous [step 1](#) is again not required to do, but only to demonstrate the full functionality of the plugin. Circle back to our plugin in My Tools tab and Press on the Add Parameters to Families icon which will prompt you to add an excel file. Locate the ParametersData.xlsx that you saved earlier and choose it as shown in [Figure 24](#).

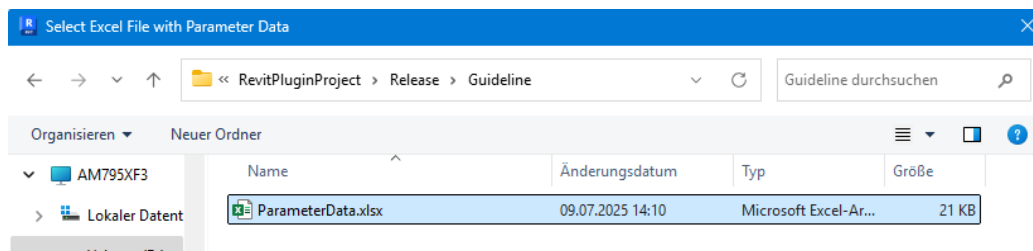


Figure 27 Plugin Prompt to Select an Excel file

After choosing the excel file, another window pops up prompting you to choose a .txt file. This time we will choose the .txt that we created earlier in the previous [step 1](#), [Figure 25](#).

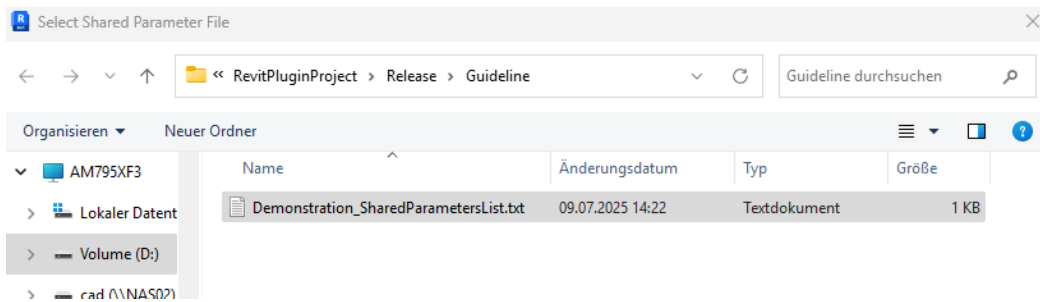


Figure 28 Choosing the Newly Created SharedParameters .txt file

If everything is successful, you should see this warning message in [Figure 26](#). We will discuss the contents of the warning message in the [More Information](#) section below, for now close it and close the plugin window that will pop after the message, we will come back to it later.

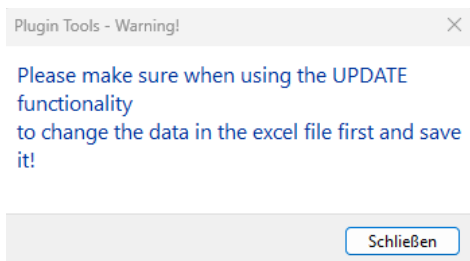


Figure 29 Warning Message

Now the data has been read from excel registered in memory and the plugin is now pointing at the shared parameters file we have chosen populating the file with the parameters in their correct datatypes under their correct groups. Open the Shared parameters (Visit [step 1](#) if unsure how) and check thoroughly that all the parameters are there under their correct groups, compare with the ParameterData.xlsx file, [Figure 27](#) shows the list populated.

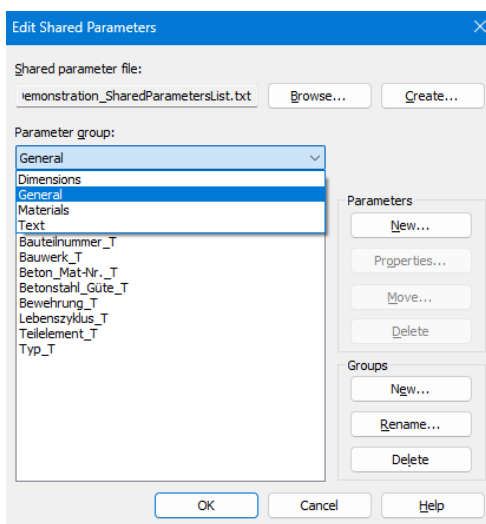
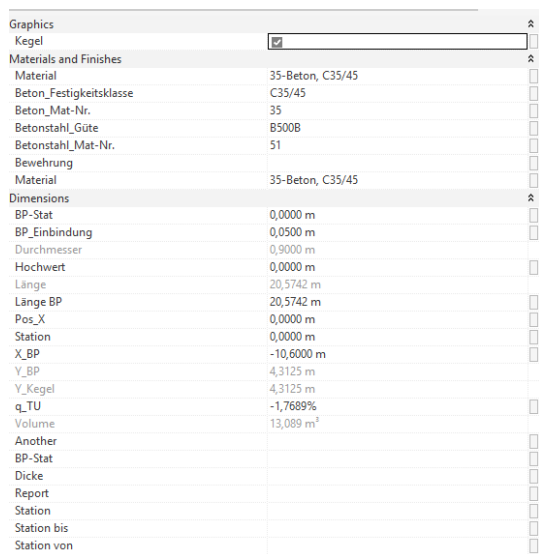


Figure 30 The shared Parameters List populated

If you have checked thoroughly and are satisfied with the result, proceed to close the window and move on to [Step 3](#).

Step 3

We will begin our demonstration on a specific Bauteil with the Bauteilnummer 142. Open Hide Elements tool and choose 142 then press Isolate. Click on any of the elements in front of you and observe the properties of this element, depending on the model you have opened you might have all the parameters and their values already in there, but as I mentioned for the sake of this guide, I have changed the names of the parameters. Based on this, my properties will look like this in [Figure 28](#).



Graphics	
Kegel	☒
Materials and Finishes	
Material	35-Beton, C35/45
Beton_Festigkeitsklasse	C35/45
Beton_Mat-Nr.	35
Betonstahl_Güte	B500B
Betonstahl_Mat-Nr.	51
Bewehrung	
Material	35-Beton, C35/45
Dimensions	
BP-Stat	0,0000 m
BP_Einbindung	0,0500 m
Durchmesser	0,9000 m
Hochwert	0,0000 m
Länge	20,5742 m
Länge BP	20,5742 m
Pos_X	0,0000 m
Station	0,0000 m
X_BP	-10,6000 m
Y_BP	4,3125 m
Y_Kegel	4,3125 m
q_TU	-1,7689%
Volume	13,089 m³
Another	
BP-Stat	
Dicke	
Report	
Station	
Station bis	
Station von	

Figure 31 Properties of One Family Instance of Bauteilnummer 142

Open the Add Parameters to Families plugin, you will see 3 buttons on the window that has popped open as shown in [Figure 29](#).

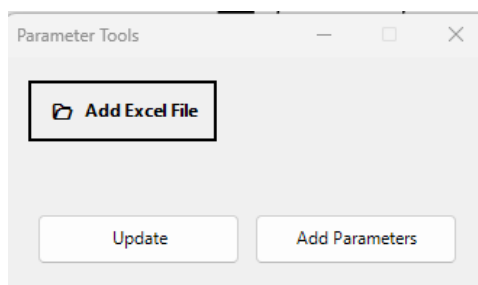


Figure 32 Main menu of the Add Parameters to Families Plugin

We will go through each of them to explain in detail the functionality of the three buttons.

Press on the Add Parameters button, this will pop up a window that contains the full list of all the Bauteilnummern that have been read from the excel file, a button to select all, a button to Clear All selection, a button to Use Selection when check marking any amount of Bauteilnummer boxes and finally a Add All button which proceeds to add all the parameters and their values accordingly. The window is shown in [Figure 30](#).

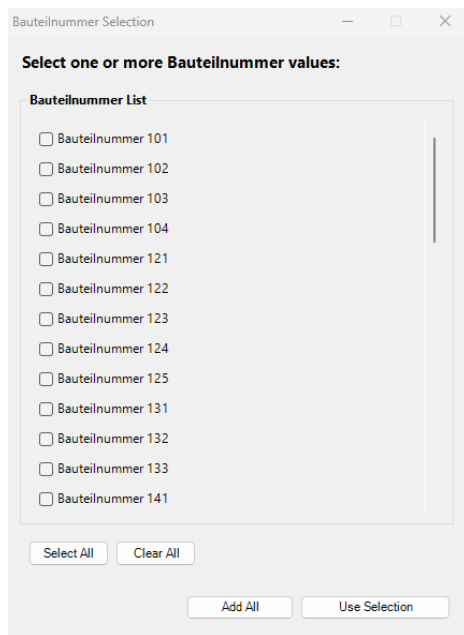


Figure 33 Selection Window

Check the box beside Bauteilnummer 142 and press Use Selection to process the information for only elements that have the Bauteilnummer 142, to demonstrate faster as the plugin would normally take around 7-10 minutes to add every parameter and every value to all the elements in the project. This long processing time will be explained in the [More Information](#) section. Proceed as shown in [Figure 31](#).

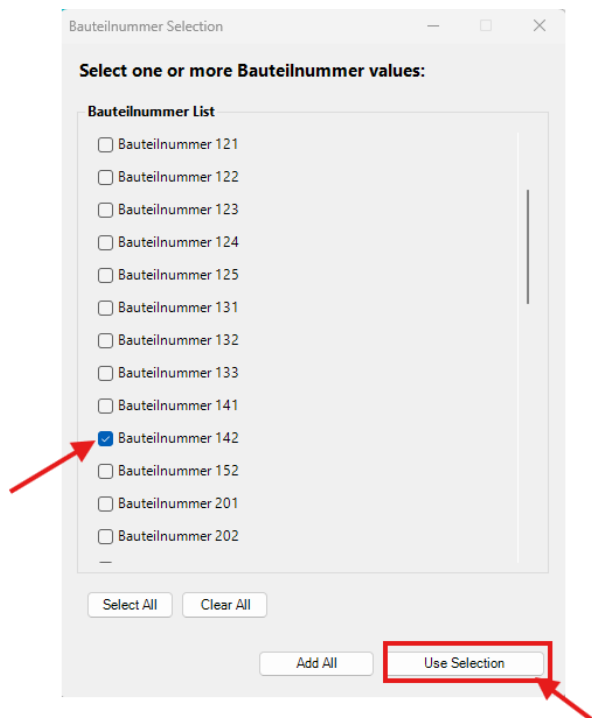


Figure 34 The selection Window to Process

The plugin will take a few seconds to processes adding the parameters to the family of the element, and if there're multiple families holding the same Bauteilnummer the plugin will process them accordingly, then proceed to add the values to those parameters. The

plugin also handles the case if the parameters exist in each element and skips it and if there are parameters that have values it will also skip it. Updating the values comes in a later stage of this guide. If the plugin ran successfully, this message in [Figure 32](#) should show a summary of how many parameters were added and how many values were applied.

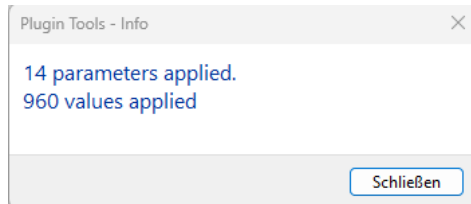


Figure 35 Summary of Parameters and Values added

Close this window and the plugin window then click on any element that has the 142 Bauteilnummer, proceed to inspect the properties window on the left, you will see that all the parameters and their values were added correctly. One can cross check with the Excel file. [Figure 33](#) and [Figure 34](#) show the result.

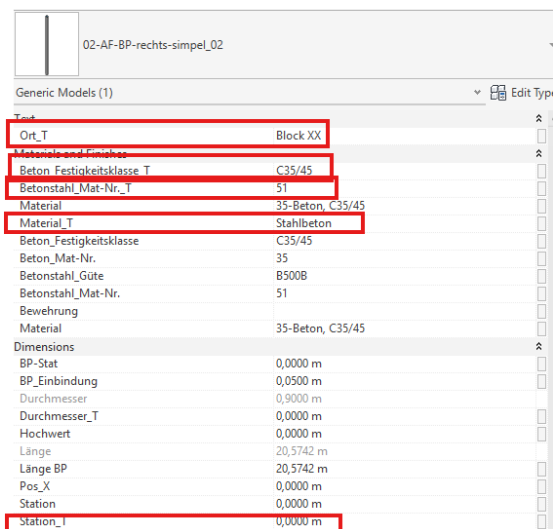


Figure 36 Result of parameters and values added

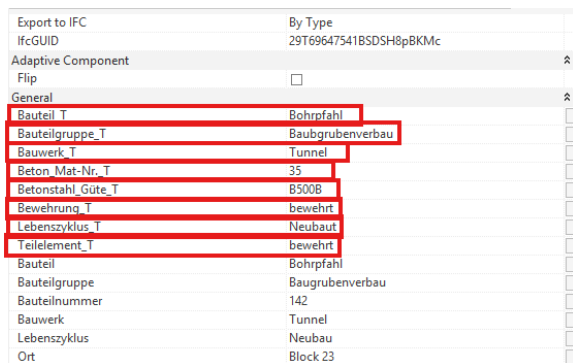


Figure 37 Rest of the Results

If the plugin hasn't added any parameters or values a message will indicate this. You will notice that parameters Station, Station bis, Station von, Ort, Breite, Durchmesser and

Dicke don't have values, that is because we ignore adding values to these parameters since they're added using the SBIM tool. You can test this on any number of elements in the project, given they have a Bauteilnummer.

The parameters that were added were added inside each of the family files from the shared parameters list to mimic the same behavior of the modeler. The values are added to each of the elements

Step 4

Next, we will discuss the Update functionality. Open the plugin again and have the excel sheet also opened, because we will start editing some values. Since we populated only the elements with the Bauteilnummer 142, we will demonstrate the update functionality on this Bauteilnummer.

Please Note: If you open Revit then open a project and the first thing you do is use the update functionality, a warning message will pop off essentially alerting the user to start editing the excel file once the plugin has ran at least once just close it, do the desired edits in the excel file then reopen the plugin and use the update functionality normally.

If you have opened the excel file, locate the row where the Bauteilnummer is 142 and change values, I will change the *Lebenszyklus_T*, *Beton_Mat-Nr._T* and *Material_T* to the following values NeubauXXX_T, 3553, Elastomer, don't forget to save the excel file. Press Update as shown in [Figure 35](#).

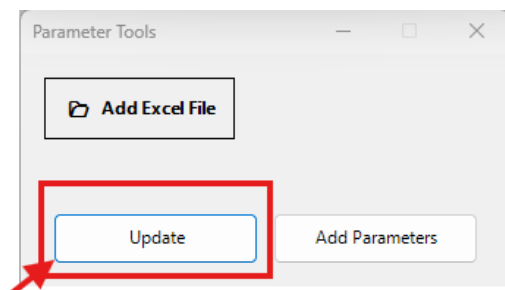


Figure 38 Update Feature

If the update was successful, you should see this message in [Figure 36](#).

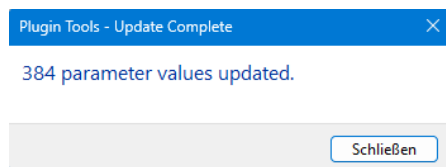


Figure 39 Summary of Updated Parameters

Finally, let us inspect the properties of one of the elements of Bauteilnummer 142. Results shown in [Figure 37](#) and [Figure 38](#).

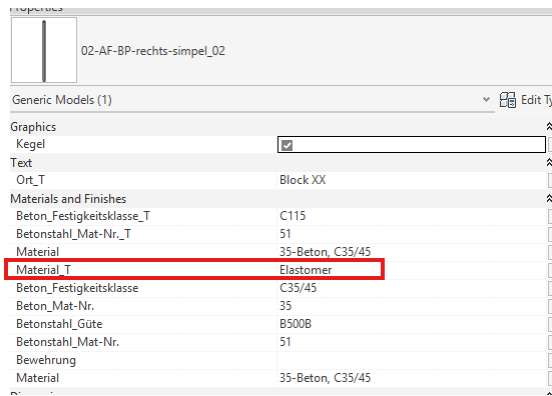


Figure 40 Result of the Update1



Figure 41 Result of the Update2

You can check all the elements of Bauteilnummer 142 to check. Let us revert what was done with a simple undo. **Do not forget to change the values in the excel sheet and saving if you are only testing.**

Step 5

Lastly, the Add Excel File button on the main menu of the plugin simply allows you to choose an excel sheet to be read and saved into memory. This newly added Excel file will be the source now and replace the data with the old one. Simply press the button shown in [Figure 39](#) and choose the desired excel sheet. **Please make sure that it is following the same structure as the excel sheet provided with the plugin.**

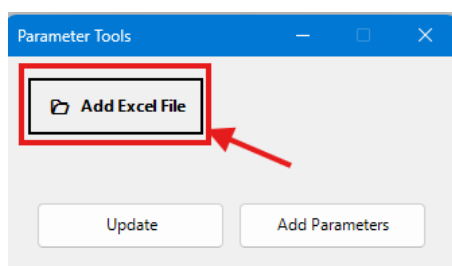


Figure 42 Add Excel File

Check For Updates Tool

The Check For Updates Tool is a convenient tool to automatically search for new updates for the plugin tools and automatically does the file updating upon closing Revit. Let us walk through the process.

The first thing that will happen when you press the Check for updates button is a message informing the user whether there's a difference in version, meaning there's an update available, or if the user is using the latest version of the Revit.Addin.2024 tools. Demonstrated below in [Figure 43](#) is the message informing the user of an available update.

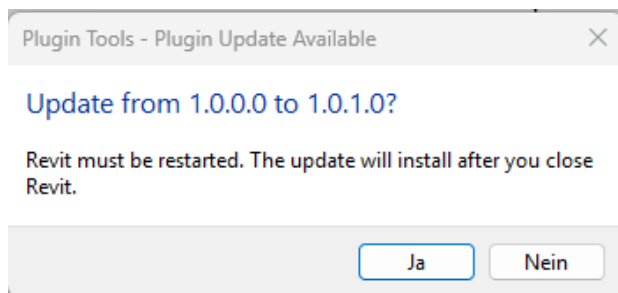


Figure 43 Available update message

The message gives information about the current version, and the new versions numbers. Choose yes if you wish to fetch the latest updates upon closing Revit, and no if you don't wish to update the tool after closing Revit.

If the user agrees to the update and message will pop up that confirms that an update will take place after Revit has closed. Shown below in [Figure 44](#).

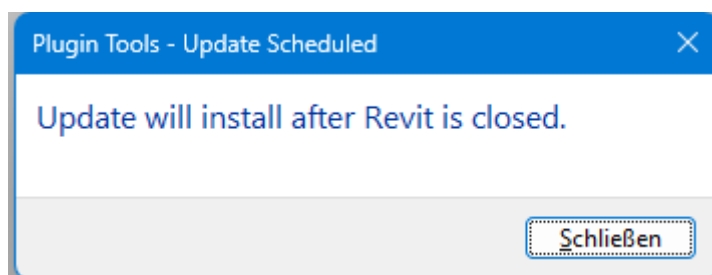


Figure 44 Confirmation Message

Proceed to close this message, then close Revit. Then after a few seconds, perhaps 2 to 5 seconds, a PowerShell Window will pop up fetching the updates from the server then proceeds to close. Unfortunately, there isn't any figure demonstration for this in the meantime.

To make sure that the update has run successfully, proceed to re-open Revit and click on the Check for Updates tool again. You should be able to see the following message in [Figure 45](#).

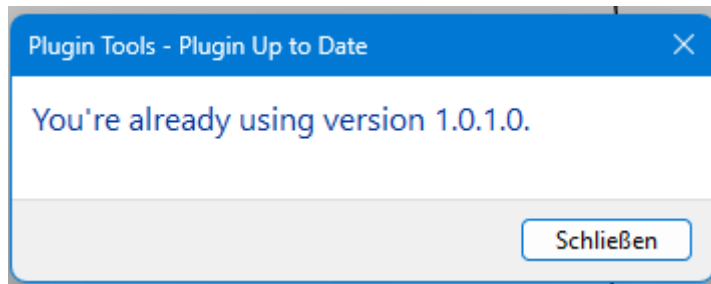


Figure 45 Successful Update

This message confirms that the user has the latest version of the Revit.Addin.2024 tools. To be more certain that the updates had taken place, check the [this Excel File](#) for a summary on what the latest version has changed/removed/added and proceed to try out the respective changes, if the behavior is identical to what is described then the update process has surely been a success.

More Information:

- The Sofistik tool that the modeler uses automatically creates one family file for each instance that is placed inside the project, hence why the plugin would take a very long time to process since it is opening each family file adding parameters into it, saving it and then loads it into the project and so on. One can imagine the amount of time it would take to open multiple family files and apply this logic.
- The contents of the warning message: Since excel is not directly connected to Revit, when Revit is closed (Session ended) the data that was read into the memory by using the plugin is automatically deleted. Following this logic, if one was to edit the excel sheet when the old information is not registered in memory, there won't be anything that could track such a change, consequently this will lead into unusual behavior when using the update functionality and might add wrong information or no information at all when updating. So, just like the warning states at the beginning of this guide/documentation, if one wishes to edit information inside the excel sheet, one should open Revit first, run the plugin once which prompts the user to choose an excel sheet, choose it then the information is now registered then editing any number of information one desires.
- If you wish to add a new column (New parameter) to the excel sheet, you can add it then do [Step 2](#) and [Step 3](#) to add the new parameter and its values to the desired Bauteils.
- Section [File Copying](#) won't be needed once a fetch new versions functionality is implemented.